# Matthew Ngoy

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## Summary

Computer science senior preparing for PhD research in AI security, focused on LLM red-teaming, jailbreak robustness, and verification of AI computation.

## Education

**Kean University** — Union, NJ

*Bachelor of Science in Computer Science (Cybersecurity Option); GPA: 3.52* — *Expected May 2027*

- Honors & Programs: NSF IUSE Scholar (2025, 2026); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WELL Program; Young Envoys Scholarship.

## Technical Skills

**Languages**: Python, Java, JavaScript, SQL, PHP, PowerShell

**Research & Security**: LLM benchmarking, adversarial testing, citation checking, OSINT, OWASP LLM Top 10, MITRE ATLAS

**Tools**: Linux (Ubuntu, Kali), Docker, Git, MySQL, SQLite, Node.js, Arduino/ESP8266, Ollama, OpenAI API

**Coursework Exposure**: TypeScript, Wireshark, Metasploit, Hashcat, Autopsy

## Experience

**Cybersecurity Research Intern** — May 2026 – Present

*Kean University Center for Cybersecurity* — *Union, NJ / Wenzhou and Beijing, China*

- Investigated nonprofit cyber incidents through OSINT for the Non-Profit Cyber Incident Repository, classifying reports by CIA-triad impact, CISA sector, and TAG Cyber taxonomy.
- Taught cryptography, programming, and IP addressing across 12 lessons at the Port Authority of NY/NJ STEaM Camp, reaching **60** high school students.

**Peer Mentor / Code Samurai** — Jan. 2025 – May 2025; Sep. 2026 – Present

*Kean University Department of Computer Science and Technology* — *Union, NJ*

- Guide students through Python, SQL, and debugging in one-on-one sessions so they can identify and explain their next steps; returned as a paid mentor after volunteering in Spring 2025.

**Secretary** — 2026

*Women in CyberSecurity (WiCyS), Kean University Chapter* — *Union, NJ*

- Coordinated guest-speaker scheduling and campus event planning for **PHD4FREE!**, developing the research and graduate-funding event from August through October 2026.
- Presented pathways into research, paid undergraduate programs, and graduate fellowships alongside Notre Dame PhD student Demetrius Hernandez, reaching **15 attendees**.

**International Youth Delegate** — Sep. 2025

*International Forum "We, the Youth" (IFWY)* — *New York, NY / Seoul, South Korea*

- Spearheaded project design and scope for SelectSolar, a proposal to divert solar panels from landfills and deliver low-cost, energy-efficient power solutions to educational and non-profit institutions.
- Selected for a fully funded fellowship at Cornell Tech, then voted 1 of 25 North American delegates for the IFWY Final Conference in Seoul, collaborating on international environmental policy and sustainable solutions.

## Projects

**LLM Red-Team Harness** | *Python, SQLite, Ollama* — June 2026

- Built an AI-assisted pipeline for prompt-injection, jailbreak, and data-leak tests, preserving judge verdicts and transcripts in a public archive with offline tests.

## Research & Presentations

**HadAgent Verification Study** | *Advisor: Dr. Boyang Li* — Sep. 2026 – Present

- Investigated whether verifier acceptance establishes model execution; directed the research questions and interpreted local tests designed and run by AI agents against a copied HadAgent verifier.
- Identified a **score-validation** gap and assessed a proposed repair through local synthetic-input tests; the repaired prototype still accepted a model claim without model execution.
- Disclosed the findings to the authors, limiting conclusions to the tested local prototype rather than deployed network behavior.

**Debate Coach LLM Evaluation** | *CRA UR2PhD* — Feb. 2026 – Apr. 2026

- Helped fine-tune and evaluate LLM debate coaches designed to challenge users' arguments, working with two teammates.
- Audited **citation reliability** with teammates and AI assistance; contributed participant interviews and presented the project at the CRA UR2PhD research showcase.

**Adaptive Learning with LLMs** | *NSF IUSE Scholars Academy* — 2025

- Prepared and checked synthetic tutoring dialogues with Node.js and monitored fine-tuning jobs; presented qualitative response comparisons as **first author** at Great Minds in STEM.

**Gen Z Housing Demographics** | *NSF IUSE Scholars Academy* — 2026

- Corrected a county/state aggregation mismatch as **sole student author**; presented the weak county-level association at Great Minds in STEM without inferring causation.